

# Suhas Suresha

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## OBJECTIVE

MSc Artificial Intelligence and Data Science student at Keele University with hands-on experience across the AI/ML lifecycle, including SQL-driven data engineering, data modelling, cloud-based model training (AWS, GCP), Generative AI, RAG pipelines, agentic workflows, LLM safety guardrails, and MLOps. Skilled in Python, SQL, real-world data integration and cleaning, with the ability to translate complex technical work into decision-ready insights for technical and business stakeholders. Fast, self-directed learner seeking roles spanning Data AI, applied machine learning, AI infrastructure, knowledge engineering, and AI strategy.

## KEY SKILLS

- **AI & ML and Programming skills:** Deep Learning, NLP, LLMs, TensorFlow, PyTorch, Scikit-learn, Agentic Systems, Model Evaluation, Python, SQL, JavaScript, Git/GitHub, Command Line, Linux/Unix Shell.
- **GenAI & LLM Engineering:** Retrieval-Augmented Generation (RAG), Vector Stores, LangGraph, Stateful Workflows, Prompt Engineering, LLM Guardrails, Input/Output Validation, JSON-Schema Enforcement, Prompt-Injection Detection, PII Detection
- **Cloud & AI Infrastructure:** AWS (EC2, SageMaker, S3) and GCP (BigQuery) for model training, hosting, data storage, scalable ML workloads and MLOps, Model Training, Model Deployment.
- **SQL & Data Engineering:** SQL, data querying, cleaning, integration, transformation, and handling large, real-world, imperfect datasets.
- **Data Modelling & Knowledge Structuring:** Relational Databases, Data Modelling, ETL, Facts, dimensions, measures, grain, hierarchies, and structuring unstructured content for analysis and AI systems.
- **Visualisation & Reporting:** GPlotly, Dash, Stakeholder Documentation, RAID Logs.

## EXPERIENCE

**Junior Data Scientist**  
*REVIORA Technologies*

*Bangalore, India*  
*Jan 2025 – Jul 2025*

- **Programming & Development:** Engineered automated data cleaning and preprocessing pipelines using SQL and Python for a 249,000-row × 18-column multi-source dataset, standardising inconsistent real-world data and reducing processing time by 20–50%, eliminating a recurring bottleneck for the research team.
- **Feature Engineering:** Designed structured data models and engineered predictive features from the cleaned dataset, improving downstream model accuracy and data reliability by 5–15 percentage points while establishing a consistent, analysis-ready data foundation for subsequent modelling and experimentation.
- **Stakeholder Reporting:** Collaborated with cross-functional research teams to translate data-quality improvements and model performance gains into clear, decision-ready reporting, enabling stakeholders across technical and non-technical functions to make more informed model-development decisions.

**Data Engineering Intern**  
*Hindustan Aeronautics Limited (HAL)*

*Bangalore, India*  
*Apr 2024 – May 2024*

- **Test Engineering:** Executed structured, data-driven test cycles on combat aircraft systems, applying systematic test planning, validation, and results analysis within a governance-led engineering programme, producing technically precise documentation in a safety-critical, zero-tolerance-for-error environment.
- **Systems testing:** Coordinated and documented test outcomes across cross-functional engineering workshops, consolidating technical findings, decisions, actions, and test evidence into traceable, audit-ready records, enabling downstream engineering teams to reliably interpret and act on results.
- **Independent Delivery:** Rapidly developed working proficiency in new technical tools, systems, and engineering workflows within a 6-week placement, progressing from initial onboarding to contributing directly to live testing workstreams with minimal supervision.

## RESEARCH PROJECTS

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### MSc Dissertation – Testing, Not Assuming: A Confound-Controlled Approach to Cognitive Load Proxy Validity in Formula 1 Telemetry

*Supervisor: Dr. Nadia Kanwal    Keele University, United Kingdom    Jun 2026 - Aug 2026*

- **Pipeline Implementation:** Built an end-to-end Python telemetry analytics pipeline using OpenF1 data, pandas, scikit-learn, XGBoost and SHAP, engineering throttle smoothness, braking consistency and sector-time variance features across 61 F1 race weekends, 28 drivers and 65,000+ laps, with confound control and driver-grouped cross-validation.
- **Predictive Modelling:** Developed and validated logistic regression, Random Forest and XGBoost classifiers, achieving a ROC-AUC of 0.740 with XGBoost, and deployed the model as a queryable reporting tool with both CLI and Tkinter interfaces for driver/race-level analysis.

### Self-Healing Retrieval-Augmented Generation & Answer Validation Pipeline

*Supervisor: Dr. Ed De Quency    Keele University, United Kingdom    Feb 2026 - Jun 2026*

- **Pipeline Implementation:** Built an end-to-end self-healing RAG pipeline over a 5,000-document corpus with 12,000 vector chunks, integrating retrieval, generation, hallucination detection, query reformulation, and fallback handling using LangGraph.
- **Self-Correction:** Designed a critic agent within a stateful, cyclical LangGraph workflow to evaluate answer grounding, automatically trigger query reformulation and re-retrieval for unsupported responses, and provide a safe “insufficient information” fallback instead of returning ungrounded answers.

### Configurable LLM Safety Guardrails & Compliance Validation Gateway

*Supervisor: Dr. Nadia Kanwal    Keele University, United Kingdom    Jun 2026 - Aug 2026*

- **Gateway Implementation:** Developed a configurable LLM middleware gateway enforcing 30 safety and compliance policies through guardrails, with automated validation across 2,000 adversarial and validation prompts covering prompt injection, jailbreaks, PII leakage, toxicity, off-topic responses, and schema violations.
- **Policy Enforcement:** Implemented input and output validation with a YAML-based policy engine, blocking unsafe requests, filtering toxic content, enforcing JSON schemas and topical relevance, and triggering automatic retries or safe fallbacks when validation failed.

### Automated CI/CD Pipeline for Machine Learning Model Deployment

*BTech Thesis*

- **CI/CD Implementation:** Built an automated CI/CD workflow integrating GitHub, Docker and cloud deployment to automate code validation, testing, containerisation and deployment of a machine learning application, reducing manual deployment steps and improving reproducibility across development cycles.

## EDUCATION

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### Keele University, United Kingdom

*2025 – Present*

*MSc Artificial Intelligence and Data Science*

- **Expected:** Distinction

### BMS Institute of Technology, India

*2022 – 2024*

*BTech Electronics and Communication*

## LEADERSHIP & EXTRACURRICULAR

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### Co-Captain, AWS Club, Keele University

*Jan 2026 - Present*

- Co-lead the university’s AWS Club, organising hands-on workshops and technical sessions on AWS services and managing membership and student engagement.

### Student Voice Representative, MSc AI & Data Science, Keele University

*Oct 2025 - Present*

- Represent the MSc AI Data Science cohort’s feedback to programme staff and faculty, raising issues and advocating for curriculum and course-experience improvements.

## RIGHT TO WORK

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UK Student visa (CAS end date September 2026); eligible for full-time employment under the Graduate visa route thereafter. Willing to undergo an enhanced DBS check and to relocate for the right role.